Greenpod Labs — Assignment 2

Computer Vision Classification of Orange Condition Categories

Dataset disclosure: The assignment asks for avocado ripening-stage classification. This project uses an orange image dataset with labels fresh, blackspot, canker, and grenning. It therefore does not solve avocado ripening classification. Confirm with the evaluator whether this substitution is acceptable.

1. Objective

Classify a whole orange image into fresh, blackspot, canker, or grenning. This is image-level classification, not object detection: the model predicts one label and does not draw bounding boxes.

2. Dataset understanding

The archive contains 1,090 readable images: 991 training and 99 test images. No unreadable images were found in the audit.

Split	blackspot	canker	fresh	grenning	Total
Train	184	179	281	347	991
Test	22	22	33	22	99

The training split is imbalanced: grenning is the largest class (347; about 35.0%) and canker the smallest (179; about 18.1%). The test set has only 22–33 images per class, so each error can noticeably change per-class recall. The audit found 150 distinct image dimensions; 776 images are 800×800. A simple rule for very small images/aspect ratios flagged zero cases, but this does not rule out mislabeled, duplicate, blurry, or unusual images.

Potential biases include background, lighting, camera, source, fruit size, and capture angle. Near-duplicates or source-specific patterns across splits could overstate real-world performance. The label grenning may mean greening and should be verified.

3. Approach 1 — HSV + HOG + SVM

Pipeline: resize → normalized HSV colour histogram → HOG edge/shape descriptors → standardize features → RBF-kernel SVM with balanced class weights. Colour histograms capture colour distribution, HOG captures edges and texture, and SVM provides a lightweight baseline suitable for a small dataset. Limitations include sensitivity to lighting/background and reliance on hand-crafted features.

4. Approach 2 — Compact CNN

A compact convolutional neural network learns spatial patterns directly from resized pixels. The training pipeline includes augmentation, class weighting, and validation-based model selection. CNNs can learn combinations of colour, texture, and spatial patterns, but training from scratch on 991 images can overfit.

5. Evaluation metrics and latest results

Both approaches were evaluated on the same held-out test set of 99 images.

Metric	SVM (HSV + HOG)	Small CNN
Accuracy	90.91%	90.91%
Macro precision	90.30%	90.62%
Macro recall	90.15%	90.91%

Metric	SVM (HSV + HOG)	Small CNN
Macro F1-score	90.03%	90.40%
Weighted F1-score	90.80%	90.77%

Accuracy gives the overall correct fraction. Precision describes the reliability of predictions for a class; recall measures how many actual class examples are found. Macro-F1 weights classes equally, while weighted-F1 accounts for class support. Confusion matrices reveal which classes are confused.

Both models achieved 90.91% test accuracy. CNN macro-F1 is slightly higher (90.40% vs. 90.03%); SVM weighted-F1 is slightly higher (90.80% vs. 90.77%). The differences are small and the test set is limited, so neither model can be declared decisively superior. CNN validation accuracy varied by epoch, peaking at 94.0% at epoch 3 before falling, which may indicate overfitting or instability.

6. Example run

Image: Datasets/Orange_Dataset/test/blackspot/b (1).jpg. Actual label: blackspot. CNN prediction: blackspot (94.77% confidence). SVM prediction: blackspot. Both predictions were correct for this example. The confidence is a model estimate, not a guarantee; one example does not establish general performance.

7. Production performance

The SVM performs image preprocessing, feature extraction, and one prediction. The compact CNN uses small resized images and can run on CPU; compatible GPU hardware may improve throughput. Production latency must be measured on target hardware including preprocessing. The 90.91% score on 99 images is not a production guarantee. Test independently collected images across devices, lighting, backgrounds, farms, varieties, and severity levels; consider confidence thresholds and human review.

8. Drawbacks

• The dataset differs from the requested avocado-ripeness task.

• The test set is small, so metric estimates are uncertain.

• Labels may be ambiguous or inconsistent, especially grenning.

• Potential source/background leakage and near-duplicates have not been ruled out.

• SVM relies on hand-crafted features; CNN may overfit the small training set.

• The project classifies whole images and does not localize fruit or lesions.

9. Improvements with more time

• Obtain the specified avocado dataset or written approval for the orange dataset.

• Verify class definitions and manually audit questionable labels.

• Check duplicates and use source/group-aware splits.

• Use stratified cross-validation and confidence intervals.

• Compare transfer learning (MobileNetV3/EfficientNet-B0) with the compact CNN.

• Tune augmentation, learning rate, class weights, and SVM parameters using validation data only.

• Evaluate on a larger external test set and measure latency, memory, calibration, and class-specific failures.

• Add a review threshold for uncertain predictions and monitor data drift.

10. Where most time was spent

Most effort was spent auditing the dataset, implementing two distinct pipelines, resolving environment/dependency issues, and evaluating both models on a held-out test set. An example inference and GUI were also prepared.

11. Reproducibility

```
pip install -r requirements.txt
python train_and_evaluate.py --data-dir "./Datasets/Orange_Dataset" --output-dir outputs --epochs 5
```

The script writes audit files, metrics, plots, saved models, and an example prediction to outputs/. Model selection and tuning should use training/validation data; the test set should be reserved for final evaluation.

Final note: This report reflects the latest run shared by the candidate. Confirm that the evaluator accepts the orange dataset substitution before final submission.